

The antibacterial and antialgal enhancement of the hydroquinone acrylamide polyurethane coating based on microphase separation

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ABSTRACT

The undesirable and inevitable adhesion of marine organisms on submerged surfaces has seriously affect the environment, economy and society, so emerging and promising strategies for antifouling are required. Here, the novel and environmental strategy of the antibacterial and antialgal materials was proposed for the application of the antifouling coating without releasing harmful substances. The environment-friendly antifouling agent, the capsaicin derivative N-(2,5-dihydroxy-4-acrylamide meth-ylbenzyl)acrylamide (PHABA), was modified to the molecular chain of the polyurethane. The best tensile strength was up to 23.5 MPa of PUP-25% and the elongation at break was 415% of PUP-25%. The excellent wear resistance (300 wear cycles) and chemical solution resistance (H₂SO₄, NaOH, and NaCl solutions) revealed the applicability of the coating. PHABA would migrate to the surface of the polyurethane coating with time and enhanced the antibacterial and antialgal properties of the coating. PUP-25% prevented more than 90% of bacterial and algal adhesion, indicating the potential application of the antifouling coating.

1. Introduction

When marine organisms (algal plants, barnacles, oysters, and other marine fouling animals) adhere the surface of submerged structures, this is named marine biofouling[1–3]. Marine biofouling has become a significant issue in the present maritime industry[4,5], such as exploiting marine resources, diminishing aquaculture, impeding subsurface transit pipelines, hastening the corrosion of ship surfaces, increasing fuel consumption and greenhouse gas emissions, and reducing fuel efficiency. The application of antifouling coating is currently the primary method to deal with marine biofouling. Traditional antifouling coatings are made of highly hazardous materials, such as tributyltin (TBT), copper [6], and silver[5], which are non-specific, hazardous to all marine organisms and the environment[7].

Environmentally friendly antifouling coatings, including tin-free self-polishing antifouling coatings[8], bionic antifouling coatings[9, 10], and low surface energy antifouling coatings[11–13], are more and more important. Ni et al.[14] reported an indole derivative-modified acrylic metal salt dendrimer antifouling coating. The addition of

indole derivatives enhanced the coating's antibacterial, anti-algae, and antifouling qualities with the resin's self-polishing characteristics. Guo et al. [15] reported an antifouling coating (PDMS-x-PU-y) by PDMS-based polyurethane and 7-amino-4-methyl coumarin. Based on the synergistic impact of the UV fluorescence antifouling capability of coumarin in daylight and the contact inhibition mechanism of the nighttime coating, the coating demonstrated excellent antifouling and fouling-release ability with a minimal amount biofouling after 4 months in sea. High preparation costs and complex synthesis processes limited the applications of the aforesaid antifouling coating in widespread practical use.

Natural compounds may naturally withstand biological contamination. Non-toxic antifouling chemicals from nature to replace chemical antifouling treatments containing heavy metals, such as copper, has become a focus of research. Capsaicin, which is extracted from natural plants like chili peppers, is the commonly utilized natural antifouling substance with the molecular formula C₁₈H₂₇NO₃[16,17]. The extraction of capsaicin can not satisfy the demands of practical applications. Based on these demands, Yu et al. [17] developed a new approach for the

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synthesis of natural products by alkylating N-hydroxymethylbenzamide with different aromatic chemicals. A series of benzamide derivatives containing capsaicin were prepared. According to this study, Jiang et al. [18] designed the environmentally beneficial Cu_2O -free bifunctional coatings by zinc acrylate resin as the self-polishing monomer and amide derivatives of capsaicin as an antifouling agent. The real sea hanging board experiment results showed the effective antifouling properties of the coatings within 9 months. Liu et al. [19] reported capsaicin-modified marine antifouling organofluorosiloxane coatings which was prepared by silicone monomer, fluoroacrylate monomer and HMBA (N-(4-hydroxy-3-methoxybenzyl)-acrylamide polymer network). The resistant to prolonged immersion in water, low surface energy, good adhesion and more than 97% antibacterial rate of this coating revealed the excellent antifouling performance. The primary base resins used in the majority of current researches are acrylic resin and hydrogel and the antifouling effect of this type of coating is good in a short time. However the quick release rate of antifouling agents results in the short service life, which limits the application in practical production.

Polyurethane resin exhibits good flexibility, wear resistance, and anti-aging qualities because it has a lot of urea and ammonia ester bonds for forming hydrogen bonds [20–22]. Hydrogen bonds increase the resin's chemical stability and adhesion [23]. It is possible to give the coating both excellent mechanical and antifouling qualities by grafting active groups with bactericidal effects onto the polyurethane matrix

resin. Therefore, the capsaicin derivative modified polyurethane antifouling coating was designed and prepared in this work. The capsaicin derivative with the terminal hydroxyl structure was successfully synthesized by the polymerization reaction of capsaicin derivative monomer (N-(2,5-dihydroxy-4-acrylamide methylbenzyl)acrylamide, PHABA) and ethylene glycol ether. The chemical structures of the compound were identified. The various solid contents of the coating were performed. The mechanical properties, hardness, static WCA, chemical stability, antibacterial and antifouling adhesion of the coating were tested. The antibacterial and antifouling mechanisms were proposed to explain its long-term static antifouling ability.

2. Experimental section

2.1. Materials

Ethylene glycol ethylene ether ($\text{C}_4\text{H}_8\text{O}_2$, AR), dimethyl formamide (DMF, AR), hydroquinone, N-hydroxymethylacrylamide, azo diisobutyronitrile (AIBN, AR), phosphate buffer (PBS), 2.5% glutaraldehyde fixative ($\text{C}_5\text{H}_8\text{O}_2$) were purchased from Shanghai Aladdin Biochemical Technology Co. Polyurethane prepolymer (the mixture of 4, 4'-diphenylmethane diisocyanate and polypropylene glycol, named as MDI-50 and PPG) were provided by Qingdao Huizhi Leading New Material Technology Co. Ltd. All chemical reagent mentioned in this work

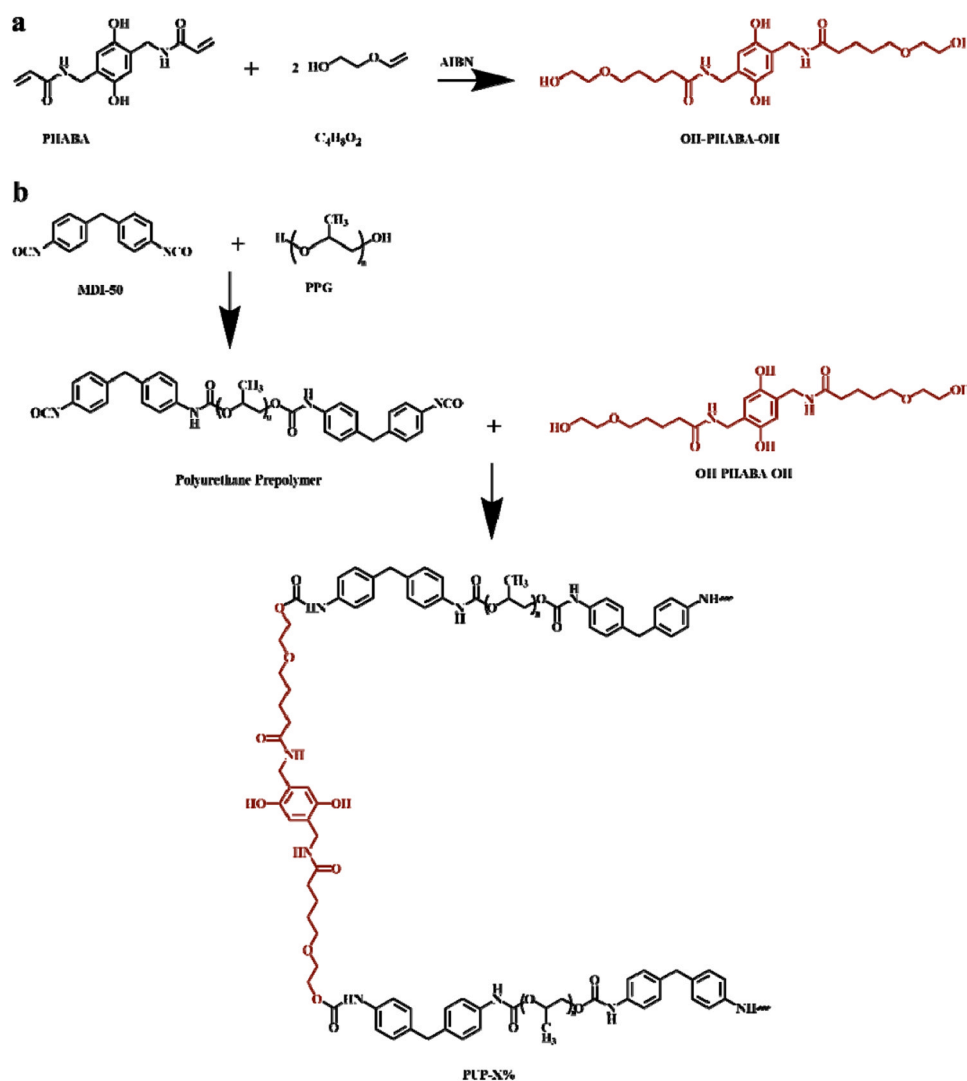


Fig. 1. The preparation process of (a) OH-PHABA-OH and (b) PUP-X%.

were used without further purification.

2.2. Synthesis of OH-PHABA-OH

N-(2,5-dihydroxy-4-acrylamide meth-ylbenzyl)acrylamide (PHABA) was synthesized according to our earlier work[24,25]. Hydroquinone and N-hydroxymethylacrylamide (molar ratio of 2:1) were added to the three-necked flask. The appropriate amount of absolute ethanol was added until all the raw materials were dissolved. The concentrated sulfuric acid was slowly added to the three-necked flask under the mechanical stirring at 35 °C. The volume of the concentrated sulfuric acid was 10% of the absolute ethanol. The crude PHABA was obtained after 8 days under the stirring at 35 °C. The crude PHABA was purified by recrystallization before being used. The chemical structure of PHABA was confirmed by FT-IR (Fig. S1), ¹H NMR(Fig. S2(a)) and ¹³C NMR (Fig. S2(b)).

OH-PHABA-OH was synthesized by PHABA and ethylene glycol ethylene ether (Fig. 1(a)). A certain quantity of dry PHABA was solved in DMF. The ethylene glycol vinyl ether (the molar ratio of PHABA to C₄H₈O₂ was 1:2) was added to the solution and kept stirring at 100 °C until the solution became transparent. Then 3–4 drops of 0.5 wt% initiator AIBN were added during the first 10 minutes of the reaction, followed by drop additions every half-hour until the reaction was terminated after 3 hours. The chemical structure of OH-PHABA-OH was confirmed by FT-IR(Fig. S1), ¹H NMR (Fig. S2(c)) and ¹³C NMR(Fig. S2 (d)).

2.3. Synthesis of PUP-X%

The prepared modified polyurethane coating was named PUP-X%, where 'X' is the solid content of PHABA (the mass fraction of PHABA in DMF during the synthesis of OH-PHABA-OH). For example, PHABA in DMF (15 wt%) was reacted with the ethylene glycol vinyl ether (the molar ratio was 1:2) to synthesize the OH-PHABA-OH, then the OH-PHABA-OH and polyurethane prepolymer were mixed to prepare the polyurethane resin, which was names as PUP-15%. As shown in Fig. 1 (b), OH-PHABA-OH was added to the polyurethane prepolymer with the mass ratio of 2.3:1. To ensure that the sample was completely self-leveling, the mixture was vacuumed and placed into the suitable mold. The polyurethane was tested after curing for 7 days.

2.4. Characterizations

The FT-IR (Spectrum-100, Perkin Elmer, USA) was utilized to analyze the chemical structure of samples. The glass transition temperatures (T_g) of the coatings were examined by Netzsch DSC 200 F3. The surface morphology and roughness of the coatings were observed by AFM (Bruker Dimension Icon). The hardness of the sample was tested by the hardness tester (LX-A of Shanghai Liuling Instrument Company), and the value of hardness was calculated an average of five times. The mechanical properties of the samples in the tension mode were measured by the universal tensile tester (INSTRON-5967, USA). The cross-head speed was 10 mm/min. The static water contact measurement (WCA) was performed by the KRUS DSA25, and the value of WCA was calculated an average of five times. Employing XRD (Rigaku SmartLab SE, Japan) to examine the crystallization of samples.

2.5. Seawater resistance test

The weighing method was used to test the seawater resistance of coatings. The samples (20 mm × 20 mm × 1 mm) were immersed in seawater at 25 °C at different times, and the value of seawater resistance capacity of coatings was calculated an average of five times. The seawater resistance capacity of coatings (w_a) was calculated as formula (1):

$$w_a = \frac{w_t - w_0}{w_0} \times 100\% \quad (1)$$

Here, the coating's mass before immersion is denoted by w₀, while its mass after immersion is indicated by w_t.

2.6. Chemical solution resistance test

The chemical solution resistance of samples was investigated by immersing 7 days in H₂SO₄, NaOH, and NaCl solutions, respectively. The WCA, hardness, and mechanical properties were discussed before and after immersion.

2.7. Antibacterial and antialgal tests

Escherichia coli (E. coli) and *Staphylococcus aureus* (S. aureus) were chosen to investigate the antibacterial properties of the coatings by plate counting method. Commence by introducing 5 mL of the selected liquid medium alongside 100 μL of a bacterial solution containing about 10⁸ CFU/mL of bacteria, into a centrifuge tube. PUP-X% are then placed within this setup. Incubation follows at a precisely controlled temperature of 37 ± 1 °C for 24 hours. After incubation, remove the sample and immerse it in 10 mL of PBS solution, ensuring a thorough shaking over 2 hours to facilitate the elution of bacteria adhered on the sample's surface into the solution. This bacterial suspension is diluted by a factor of 10³. These plates are then incubated at the constant 37 °C for 24 hours. Post-incubation, the colony count is determined, allowing for the calculation of the antibacterial activity (E_a) by comparing the colony formation in the sample plates against that in the control (blank) plates. The inhibition rate E_a was calculated as formula (2):

$$E_a = \left[\frac{N_a - N_m}{N_a} \right] \times 100\% \quad (2)$$

N_a is the number of microorganism colonies on the control tube and N_m is the number of microorganism colonies on the tube with coating.

The morphology of bacterial attachment on the surface of the coating was observed by SEM after the coating was cultured with bacteria for 48 h.

Diatoms were used as target pollutants in this experiment. 200 mL of natural seawater and 20 mL of diatom species were taken in a beaker, stirred well, and placed in a light incubator at 20 °C for 7 days. The initial concentrations of benthic diatom were above 2 × 10⁵ cells/mL. The coated specimen was placed flat in a 500 mL beaker, and the filtered and sterilised seawater and benthic diatoms were added and left to soak for 7days. The experimental light day/night time ratio was controlled between 12 h:12 h, the water temperature was controlled between (20 ± 1) °C.

2.8. Statistical analysis

In this investigation, all data are presented as the mean ± standard deviation (SD). The sample size (n) for each statistical test is explicitly documented. The threshold for statistical significance across all analyses was established at a p-value of < 0.05, with values below this threshold deemed to signify statistical significance. For the experiments assessing attachment, enumeration methods were employed to quantify the adhesion of bacteria and algae, ensuring each test for bacterial and algal species was replicated a minimum of three times (n ≥ 3) to guarantee the results' reliability and reproducibility. Statistical analyses and the generation of graphical representations were executed using Origin 2021.

3. Results and discussion

The optical photographs of all coating at 2 days and 7 days were taken in Fig. 2(a) and (b), respectively. The surface morphology of all

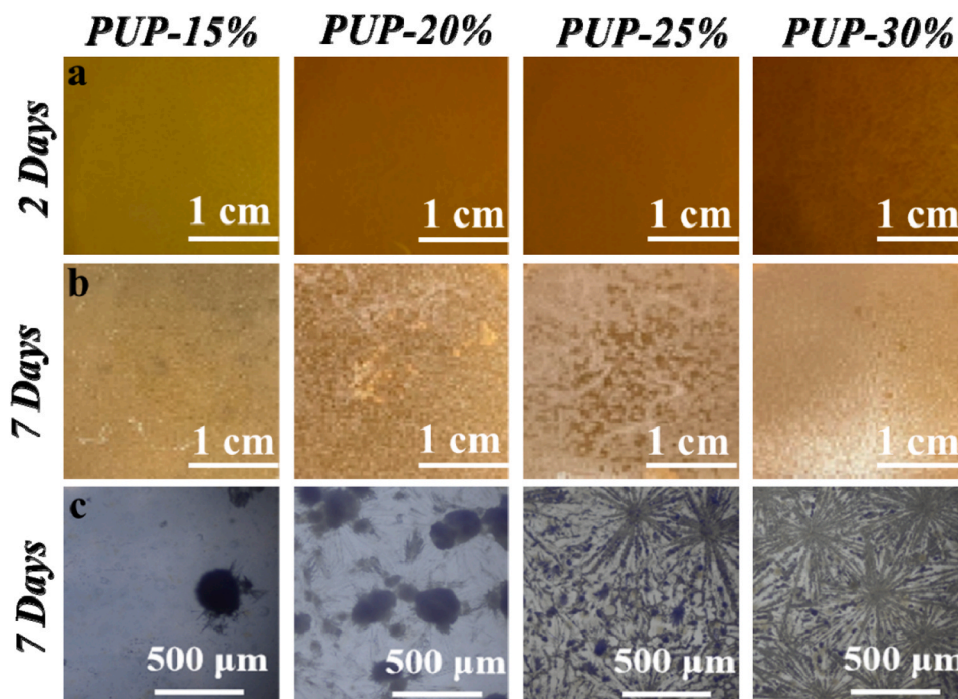


Fig. 2. Optical photographs of PUP-X% (a) at 2 days and (b) at 7 days, (c) microscope images of PUP-X% at 7 days.

coatings showed relatively flat at 2 days. After 7 days, the white substances were clear observed on the surface of all coatings and the amount of the white substances were more and more as the increase of the PHABA, as shown in Fig. 2(b). The microscopy images of the white substances on the surface of all coating at 7 days were shown in Fig. 2(c). The white substances were globular when the content of PHABA was below 20%. And the needle-like structure appeared when the content of PHABA was over 25%. The optical photographs and microscope images of all coatings at 14 days were observed and shown in Fig.S3(a) and (b), respectively. The white substances were clear observed on the surfaces of all coatings with time. This phenomenon might be due to the microphase separation of polyurethanes. The polyols as the soft segments of polyurethane are stretchable chains with the low glass transition temperature (T_g). The OH-PHABA-OH as the chain extenders with isocyanates are formed the hard segments of polyurethane with high T_g . The thermodynamic incompatibility between the soft and hard segments induces the formation of distinct phase domains, as we all know, the microphase separation[26–28]. PHABA has been proven the excellent antibacterial performances[25,29]. The white substances were the spatial aggregation of the hard segment and the amount of PHABA was increased on the surface as the aggregation of the hard segment to improve the antibacterial and antialgal of polyurethane. In further, the

chemical structure of white substances was confirmed by FT-IR (Fig. 8 (a)) to prove the aggregation of PHABA on the surface of polyurethane coating.

To further observe the microphase separation of polyurethane, SEM images of PUP-X% at 14 days on surface and the cross-section are shown in Fig.S4 (a) and (b), respectively. The needle-like cluster appeared on the surfaces of all the coating, indicating that the hard segment with PHABA structure would gradually migrate to the coating surface through microphase separation. A small amount needle-like cluster was observed in the cross-section of PUP-20%. And the island-like cluster gathered in the interior of the coating from the cross-section when the content of PHABA over 25%. The degree of microphase separation in the interior of the coating increased as the content of PHABA might improve the mechanical properties of polyurethane coatings.

The surface roughness of all coatings was investigated by AFM and the results are showed in Fig. 3. From the Fig. 3, it's clearly to be seen that the surface roughness increased along with the increasing solid content of PHABA, because that the microphase separation resulted the increasing of coating roughness. Surface chemical property has been reported to play a crucial role in the wetting behavior and surface morphology of coatings[30]. And the migration of the polar groups of PHABA to the surface as the increase of time, improving the resistance to

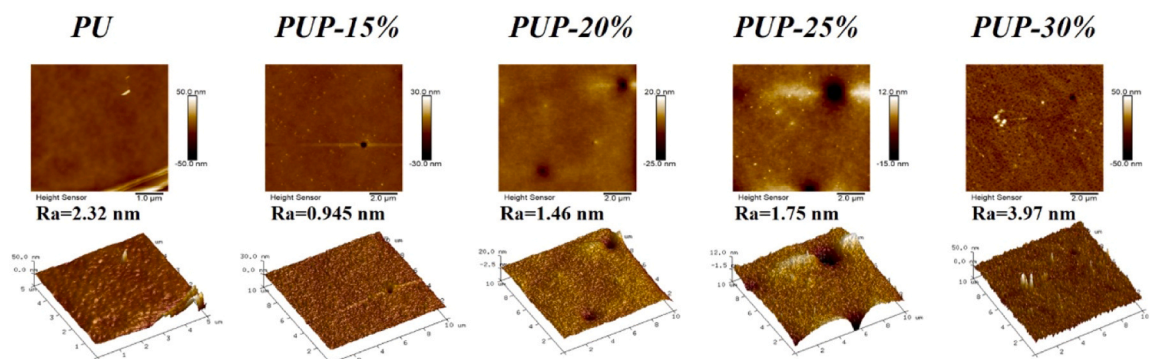


Fig. 3. The AFM images of PUP-X% with roughness.

microbial adhesion and reducing the likelihood of the organisms on the surface.

The WCA was used to characterize the microphase separation ability of the hydrophilic polymer chain to the surface. Fig. 4(a) shows the WCA results of the PUP-X% coatings before and after submerged in water for 7 days. The values of WCA of PU, PUP-15%, PUP-20%, PUP-25% and PUP-30% were $81.7^\circ \pm 1.5^\circ$, $70.3^\circ \pm 1.5^\circ$, $61.7^\circ \pm 2.1^\circ$, $52.3^\circ \pm 1.5^\circ$ and $43^\circ \pm 3^\circ$, respectively. The decrease of WCA grew progressively as the PHABA content rose, indicating that the hydrophilicity of the coating enhanced. The WCA values of PUP-X% coatings immersed in water for different days are shown in Fig. 4(b). As the increasing of time, the WCA values of all samples decreased and tended to stabilize until 6 days. These results illustrated that the action of microphase separation led to the migration of the polar groups of PHABA to the surface even in water. The hydrophilic groups on the surface could cause the water molecules to form the hydration shell on the coating's surface, preventing organic molecules and marine life from coming into direct contact with the material [31, 32]. It is advantageous to withstand protein adsorption and antifouling, indicating the potential application of ship coating in the practice of seawater.

The resistance water of the PUP-X% were investigated to confirm the coatings' dependability (Fig. 4(c)). The addition of PHABA improved the hydrophilic chain segment of the coating. The water molecule was easy to enter the chemical structure of the coating, resulting in the swelling of the coating. The cross-linking mesh structure of the coating resisted further dissolution, thus, the swelling ratio of the coating stopped increasing until 5 days. The swelling ratio of PUP-20% was the lowest among all the coatings, demonstrating the resistance water of PUP-20% is the best.

Fig. 4(d) shows the DSC curves. The T_g values of PUP-15%, PUP-20%, PUP-25% and PUP-30% were -18.53°C , -16.62°C , -15.72°C and -8.47°C , respectively. The OH-PHABA-OH had benzene ring and the polar group, blocking the movement of the polymer chain. The

movement of polymer chain required more energy to open the hydrogen bonds formed by polar group and break the hindrance of the rigid segment formed by benzene ring. Therefore, as the content of PHABA increased, the values of T_g increased. Moreover, the increasing of T_g might enhance the polymer's internal stiffness [33].

Wear resistance is one of the important indicators of coating application. Fig. 5(a) shows the wear resistance results of PUP-X%. After 300 wear cycles, the surface of the coatings exhibited no obvious damage from the micrograph. The WCA of the coatings before and after the wear test were shown in Fig. 5(b). The hydrophilic properties of the coating were decreased after rubbing 300 cycles, revealing the polar group on the surface of the coating was removed a degree. However, the PUP-20%, PUP-25%, and PUP-30% still kept hydrophilic properties, indicating a lot of the polar groups were kept on the surface after the wear test. Within the polyurethane resin, the hard and soft segments experience microphase separation. The conglomerated areas of hard segments confer wear resistance upon the coating. Furthermore, the surface of the coating, enriched with PHABA, afforded the physical protection and mitigated the direct wear and damage of the polyurethane surface upon exposure to external forces. The excellent wear resistance of coating implied the potential for practical applications.

The stress-strain curves of the PUP-X% are described in Fig. 5(c). The tensile strength of PU, PUP-15%, PUP-20%, PUP-25%, and PUP-30% were 3.1 MPa, 19.4 MPa, 23.3 MPa, 23.6 MPa, and 22.5 MPa, respectively. The tensile strength of PUP-X% was roughly 6–8 times more than that of the PU. The elongation at break of PU, PUP-15%, PUP-20%, PUP-25% and PUP-30% were 321.6%, 219.6%, 390.8%, 413.3%, and 319.5%, respectively. This is because the benzene ring and phenolic hydroxyl group in the chemical structure of OH-PHABA-OH easily formed π - π bonds and hydrogen bonds. The benzene ring of OH-PHABA-OH as the rigid structure improved the tensile strength of the coating, which was consistent with the thermodynamic results. When PUP-X% was stretched, the dynamic non-covalent, such as π - π bonds and

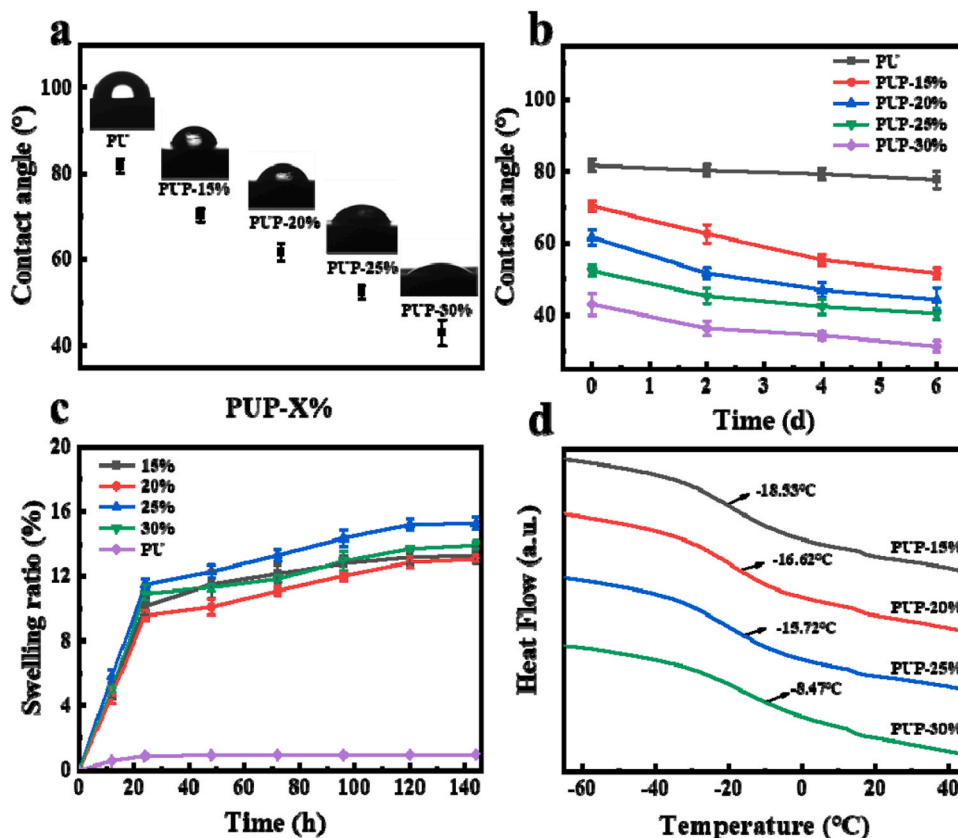


Fig. 4. (a) The WCA of PUP-X%, (b) the variation of WCA of PUP-X% under different immersion days, (c) swelling ratio, and (d) DSC curves of the PUP-X%.

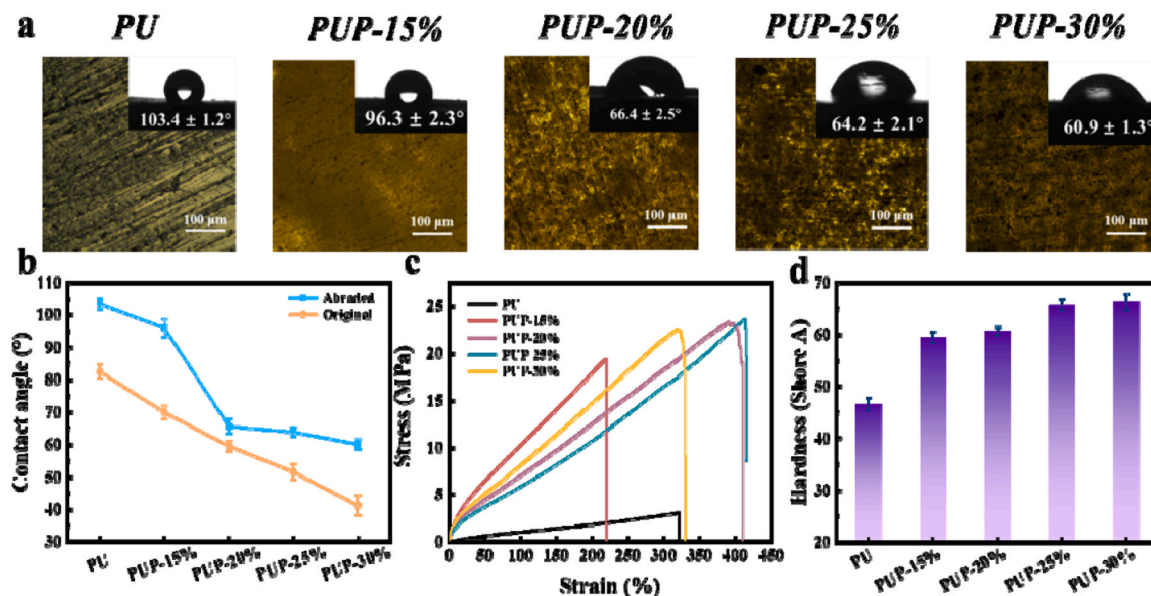


Fig. 5. (a) The fluorescent microscope images and WCA of PUP-X% after 300 wear cycles, (b) the variation of WCA, (c) curves of stress-strain, and (d) hardness values of PUP-X%.

hydrogen bonds, was broken and regrouped again, causing the elongation at the break of PUP-X% to improve to a certain extent. The benzene ring structure of OH-PHABA-OH was easily formed π - π bonds, improving the tensile strength of polyurethane coating but also impeding the mobility of polyurethane molecular chain. Consequently, the elongation at the break of PUP-30% reduced. Moreover, when the content of PHABA was excessive, the presence of unreacted PHABA caused excessive microphase separation, which diminished the tensile strength[34], and elongation at the break of PUP-30%. Overall, PUP-X% exhibited

excellent tensile properties compared to recently reported polyurethane antifouling coatings (2–12 MPa)[23,35].

Shore hardness A of PUP-X% are shown in Fig. 5(d). The hardness of PU, PUP-15%, PUP-20%, PUP-25%, and PUP-30% is 46.5 A, 59.4 A, 60.6 A, 65.6 A and 66.2 A, respectively. The addition of the OH-PHABA-OH improved the hardness from 46.5 A of PU to 66.2 A of PUP-30%. The content of PHABA increased which revealed both benzene group and polar group increase. Both the rigid structure and hydrogen bonds made the coating harden, and the change of hard results were consistent with

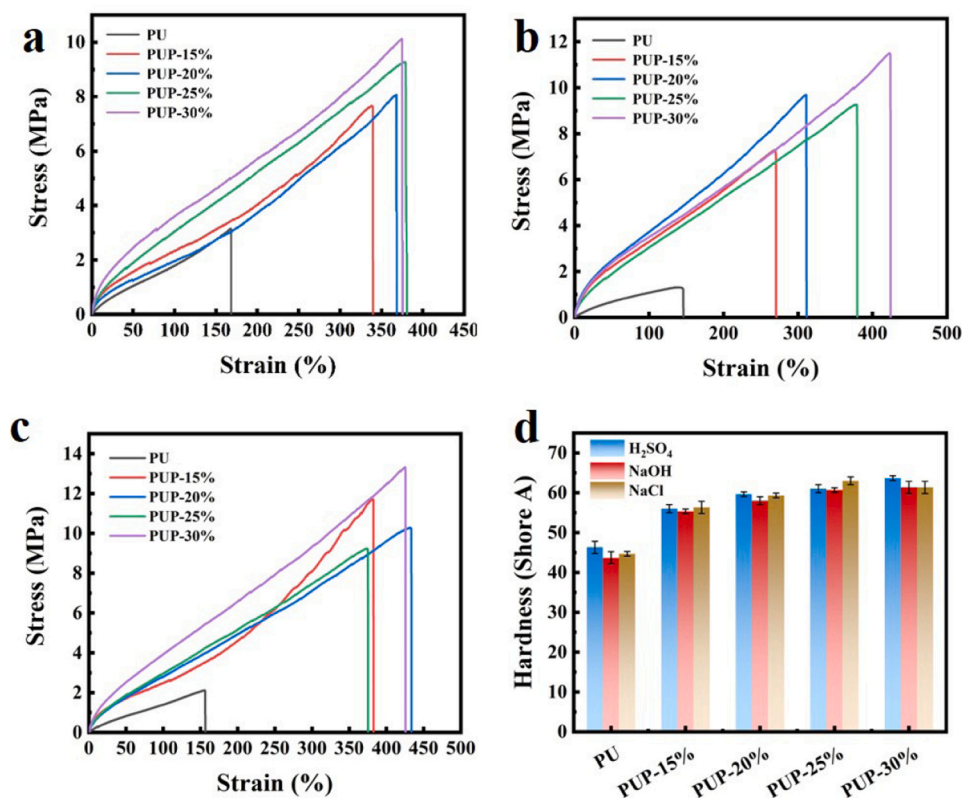


Fig. 6. (a-c) Stress-strain curves and (d) hardness of PUP-X% after 7 days of immersion in H₂SO₄, NaOH, and NaCl solutions.

the tensile strength of the coatings.

The resistance of chemical solution is one of the key properties for the application of coating. After being immersed in H_2SO_4 solution for 7 days, the tensile strength of PU (3.1 MPa), PUP-15%(7.6 MPa), PUP-20%(8.1 MPa), PUP-25%(9.3 MPa) and PUP-30%(10.1 MPa) were decreased in Fig. 6(a). The elongation at break of PU, PUP-15%, PUP-20%, PUP-25% and PUP-30% were 167.8%, 338.6%, 367.7%, 378.10%, and 374.6%, respectively. Compared with Fig. 5(c), both the tensile strength and the elongation at the break of PU decreased, indicating H_2SO_4 solution destroyed the mechanical properties of PU. Though the tensile strength of PUP-X% were decreased after being treated in H_2SO_4 solution for 7 days, the tensile strength of PUP-25% and PUP-30% still kept over 10 MPa, proving the excellent resistance of H_2SO_4 solution. Meanwhile, the negligible reduction of the elongation at break for PUP-

25% was achieved after immersing in H_2SO_4 solution for 7 days, also suggesting the excellent resistance of H_2SO_4 solution for PUP-25%. Similar results and patterns appeared in Fig. 6(c) and (d), revealing the excellent resistance of NaOH solution and NaCl solution for PUP-25%. Fig. 6(d) shows the hardness of the PUP-X% after being immersed in different chemical solutions. The negligible reduction of hardness before and after being immersed in different chemical solutions suggested that the coatings retain the excellent resistance of H_2SO_4 solution, NaOH solution, and NaCl solution. The introduction of OH-PHABA-OH into the polyurethane chemical structure, was hardly affect the excellent chemical resistance of polyurethane itself. Therefore, the PUP-X% was nearly impervious to infiltration and degradation by chemical solution, thereby safeguarding the coating against the chemical solution.

As a potential application in antifouling coatings, the research of

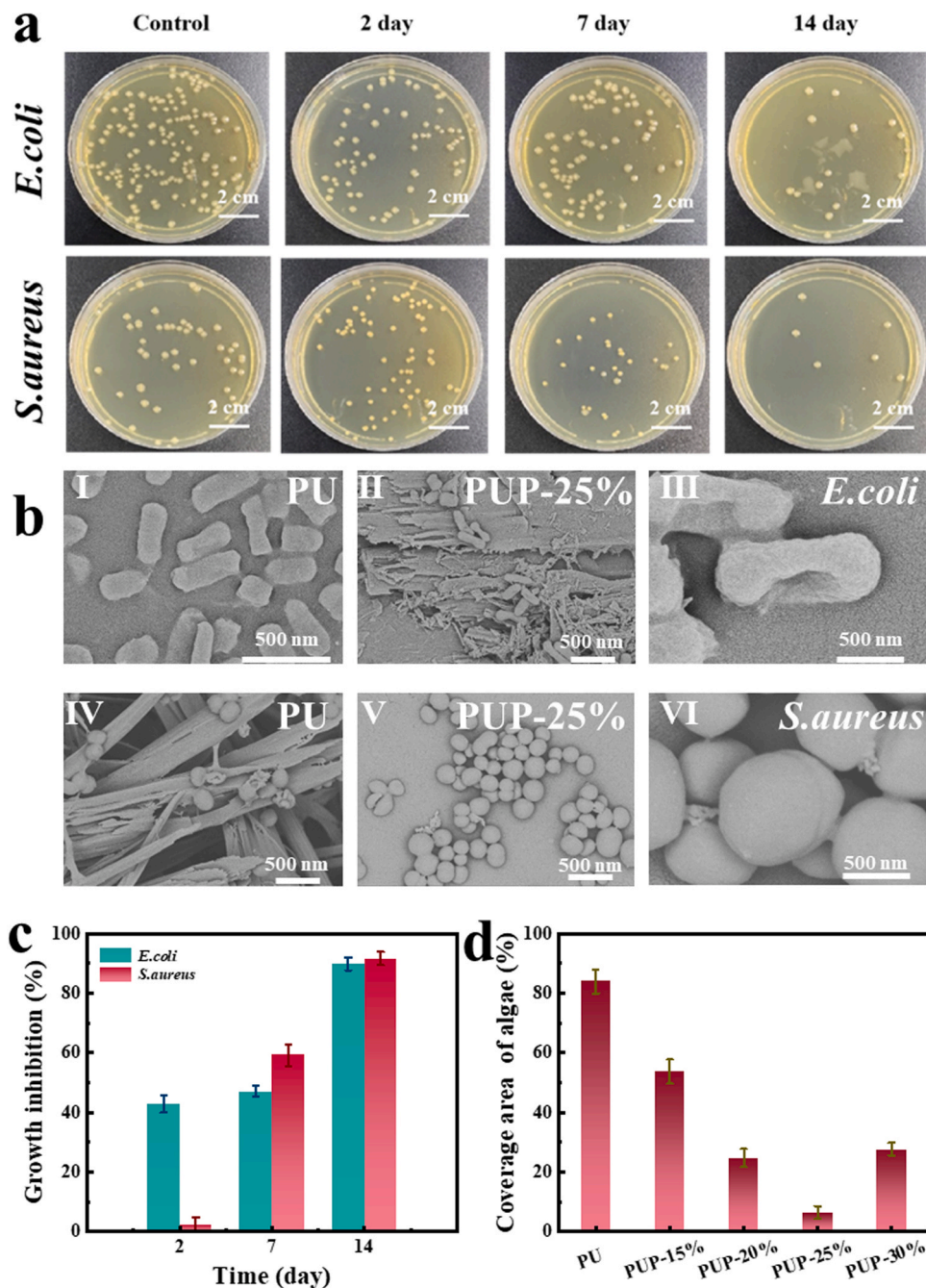


Fig. 7. (a) Antimicrobial properties of PUP-25% at 2, 7 and 14 days, (b) SEM of bacteria on the surface after 14 days of curing of PU (I and IV) and PUP-25% (II and V), and the enlarged images of bacteria on the surface of PUP-25 (III and VI), (c) growth inhibition of bacteria, and (d) coverage area of algae of PUP-25%.

bacterial adhesion on the surface of the coating is one of the necessary characterized methods. *S. aureus* and *E. coli* were chosen as the representatives of gram-positive bacteria and gram-negative bacteria, respectively. The antibacterial properties of the PUP-25% at 2, 7, and 14 days after curing are shown in Fig. 7(a). PU as controls showed a high density of bacterial adherence. The antibacterial effect of PUP-25% was better and better with the extension of curing time as shown in Fig. 7(a). The growth inhibition rate was calculated and shown in Fig. 7(c). The growth inhibition rate of PUP-25% for *E. coli* was 42%, 47%, and 89% after 2, 7 days and 14 days, respectively. Although PUP-25% shows weaker antibacterial property for *S. aureus* within 2 days, the growth inhibition increased to 59% and 91% for *S. aureus* after 7 days and 14 days, respectively. The improvement of growth inhibition was the result of the migration of PHABA to the surface of PUP-25% with the extension of time. As previously mentioned, the white substances were presented on the surface at 14 days more than that at 7 days, indicating that the amount of PHABA on the surface of coating was more and more and the antibacterial property enhanced with time. The antimicrobial and antialgal properties of PUP-25% were superior to those of recently reported antifouling coatings with capsaicin derivatives as antifouling agents[36]. Fig. 7(d) shows the morphology of attached *E. coli* and *S. aureus* on the surface of coatings. Compared with the surface of PU, the cell membrane of *E. coli* and *S. aureus* on the surface of PUP-25% had entirely collapsed and the interior cytoplasm spilled out. The antibacterial results indicated that PUP-25% destroyed the bacterial cell membrane structure.

The algae settlement behavior on the surface of the coating is also one of the necessary characterization methods for the application of antifouling. The algae growth on the surface of different coatings was observed by the fluorescent microscope (Fig. S5). As the content of PHABA increased, the amounts of algae on the coatings' surface became less and less. After cleaning with water (0.1 MPa), the algae on the surface were easily removed, revealing that the algae anchored unfirmly. The algae coverage area rates of the coatings were calculated and shown in Fig. 7(d). The algae covered area rate of PU, PUP-15%, PUP-20%, PUP-25%, and PUP-30% was 84%, 53.7%, 24.7%, 6.3%, and 27.7%, respectively. The algae covered area rate of PUP-25% decreased

from 84% (PU) to 6.3%, indicating the outstanding algal resistance of PUP-25%. The PUP-30% exhibited more algae than the PUP-25%. It's probably because the excessive PHABA increased the roughness of the coating and provided more spots for algae to attach, increasing the difficulty of removal.

The surface of PUP-X% after curing for 7 days was observed in the obvious white substances. To further provide evidence of the migration of PHABA to the surface of all coating, the white substances on the surface of PUP-25% after curing for 7 days was examined by FT-IR (Fig. 8(a)). The peaks at 3354 cm^{-1} , 3265 cm^{-1} , 1650 cm^{-1} , 1527 cm^{-1} , and 1251 cm^{-1} belonged to the stretching vibration of O-H, N-H, C=O (amide) and phenol structure C-O, respectively. The white substances on the surface of the PUP-25% has the same characteristic peaks as PHABA, proving the migration of PHABA to the surface of PUP-25%. The excellent antibacterial and algal resistance properties of PUP-25% were also the result of the migration of PHABA to the surface.

Fig. 8(b) shows the antibacterial and antialgal mechanism of PUP-25%. The thermodynamic incompatibility between the soft and hard segments of polyurethane lead to the microphase separation. The OH-PHABA-OH with antifouling property, as the chain extenders, with isocyanates are formed the hard segments of polyurethane, as red bar shown in Fig. 8(b). The polyols as the soft segments of polyurethane are stretchable chains, which provide the mobility of the polyurethane molecular chain. The spatial aggregation of the hard segment resulted the amount of PHABA was increased on the surface of coating. The phenolic hydroxyl group of PHABA caused the damage to the cell structure of microbials[25,37], thereby leading to the demise of bacteria or algae. In water environment, the hydration layer formed on the surface of the coating as the migration of PHABA groups to the surface, prevented the organisms from adhering to the surface of the coating. These two aspects worked together synergistically with each other to promote the antimicrobial and antialgal properties of PUP-25%.

4. Conclusion

The terminal hydroxyl structure with capsaicin derivative monomer (PHABA) was formed by polymerization and was introduced into

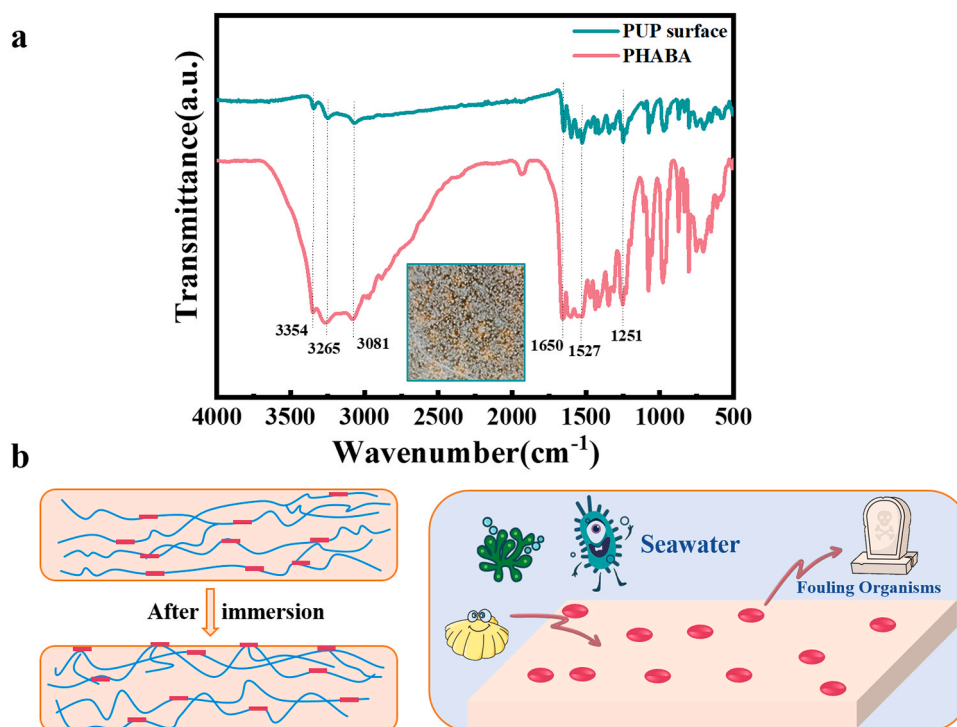


Fig. 8. (a) FT-IR of substance on the surface of PUP-25% after 7 days of curing, and (b) the antifouling mechanism of PUP-X%.

polyurethane to design a series of modified polyurethane coatings. The benzene ring and phenolic hydroxyl group in the chemical structure of PHABA easily formed π - π bonds and hydrogen bonds. The benzene ring of PHABA as the rigid structure improved the tensile strength of the coating. The dynamic non-covalent caused the elongation at the break of PUP-X% to improve to a certain extent. PUP-25% exhibited a tensile strength of 23.6 MPa and elongation at a break of 413.32%. After 14 days of curing, PUP-25% inhibited *E. coli*(89%) and *S. aureus*(91%) with the high antibacterials effect and the algae coverage area of PUP-25% was as low as 6.3%. The excellent chemical solution resistance, and antibacterial and antialgal of PUP-25% provide the potential application for polyurethane as an antifouling coating.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.colsurfb.2024.113962](https://doi.org/10.1016/j.colsurfb.2024.113962).

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